AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-19 | Entry window 19 October 2026 (full day) | 18 records screened in scope

18 RECORDS IN SCOPE (70 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

5 EFFECT ESTIMATES WITH A CI

1 TIER-A RECORDS

6 RECORDS CARRIED ON METADATA ALONE

Signal of the day

In 131,346 Guangdong schoolchildren, mass-normalised oxidative potential explains allergy-related symptoms more consistently than $PM_{2.5}$ mass — and carries up to 36 % of a constituent's association.

Six cities, ages 6–18; DTT oxidative potential as volume- (DTTv) and mass-normalised (DTTm) metrics alongside $PM_{2.5}$ mass, metals, carbonaceous fractions and ions. Mass and OP were both positively associated with rhinitis, rhinoconjunctivitis and eczema symptoms; DTTm gave lower AIC/BIC and survived two-pollutant models best. Mixture and mediation analyses put the largest indirect share at **36.2 %** (OC → DTTm → rhinoconjunctivitis), with Co, Cu and Hg also partly routed through OP. **Where it is weak:** cross-sectional symptom questionnaires, no effect sizes in the abstract, and OP assigned from a sparse filter network — a better fit may reflect smoother exposure error as much as biology. **Why it matters here:** optical low-cost sensors report mass only; if OP is the health-relevant axis, mass networks need a co-located speciation or OP tier to be interpretable for allergic endpoints.

What changed today

- **A PM null built on interpolation.** Korean stroke registry, n=27,340: post-discharge district PM_{10} and $PM_{2.5}$ (IDW) unrelated to recurrence; only SO_2 showed a graded association (Q4 vs Q1 HR **1.19**, 0.94–1.50), concentrated in harbour cities.
- **Secondary inorganic components and cognition.** CLHLS: per 10 $\mu g\,m^{-3}$ HR 1.005 for $PM_{2.5}$ but 1.030 (sulphate) and 1.033 (ammonium); a favourable lifestyle halved risk and mediated ∼9 % of the pollutant association.
- **Traffic counts do not substitute for monitors.** Bosphorus bridges, 17,520 h: a traffic-only proxy gave negative R^2 for every pollutant; ML with meteorology and autoregressive terms reached R^2 ≈0.77 — mostly persistence.
- **Component-resolved assimilation.** CALIOP–MODIS species extinction assimilated by LETKF in WRF-Chem recovers elevated free-tropospheric BC plumes over South Asia that the free model misses.
- **One record held.** *Foehn*, a generative km-scale global $PM_{2.5}$ forecaster (arXiv 2610.25941, tier A), was admitted but its DataCite DOI does not yet resolve; it is authored and held for the next issue.

Window continuity

The 18 October issue closed with `last_entry_date` at 18 October. This issue collects **19 October in full**, continuous with it. Built on the run that started 23:0x CDT on 19 October (04:0x UTC 20 October), i.e. after the 22:00 cut-off, so the day is complete. 20 October is due on the next run at or after 22:00 CDT.

Corpus analytics

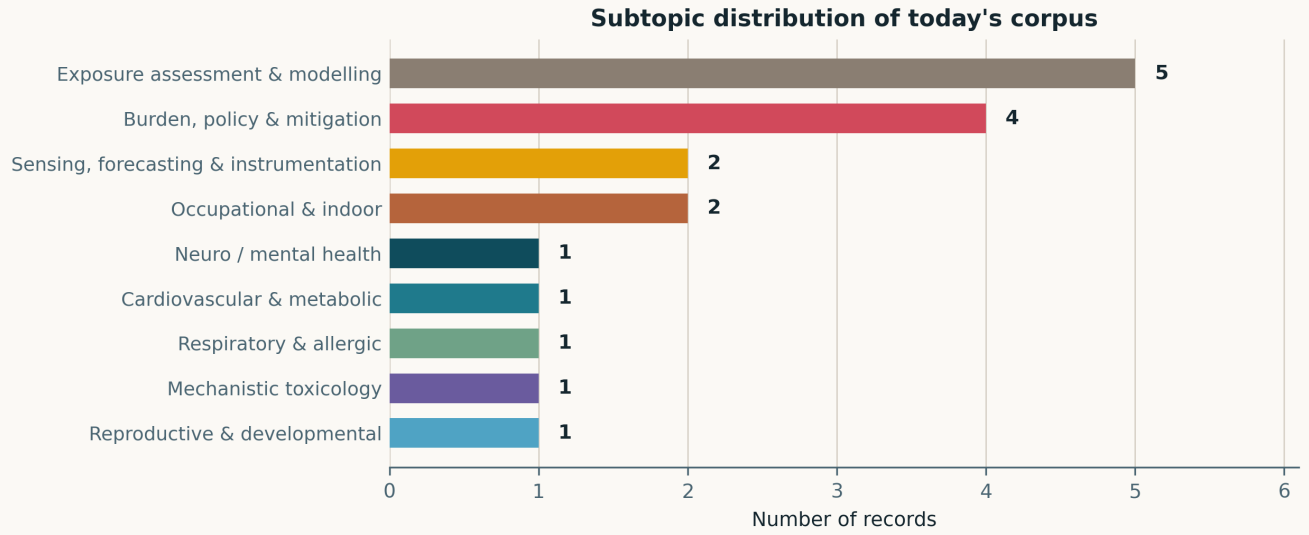


Fig. Subtopic assignment of the 18 in-scope records across nine bands. Exposure assessment (5) and burden/mitigation (4) lead; every health band holds a single record; other clinical endpoints is empty.

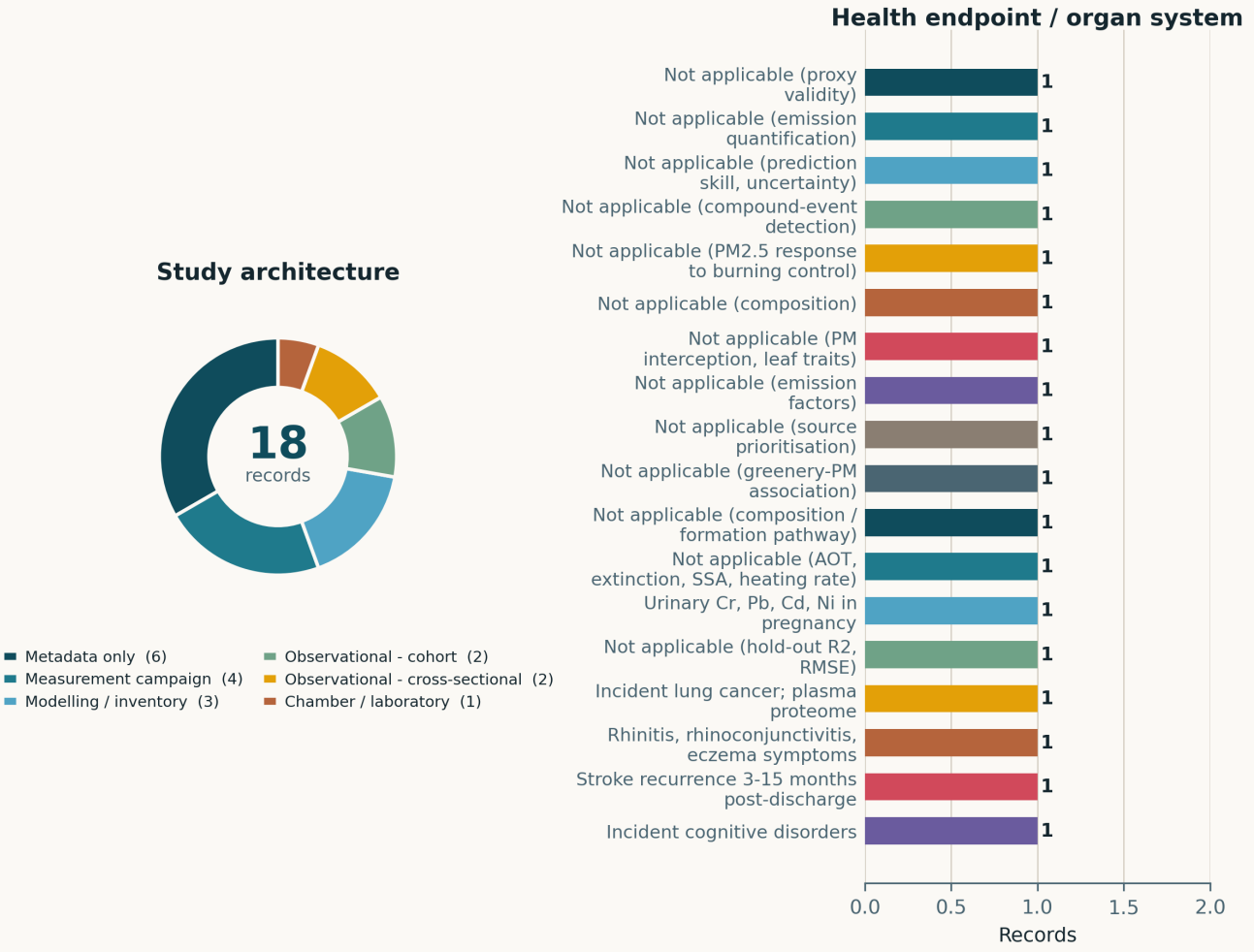


Fig. Left — architecture: two cohorts, two cross-sectional, three modelling/inventory, four measurement, one laboratory; **metadata-only 6 (33 %)**, all Elsevier/T&F titles without an indexed abstract on day 0. Right — endpoints: cognition, stroke recurrence, allergy symptoms, lung cancer proteome, urinary metals.

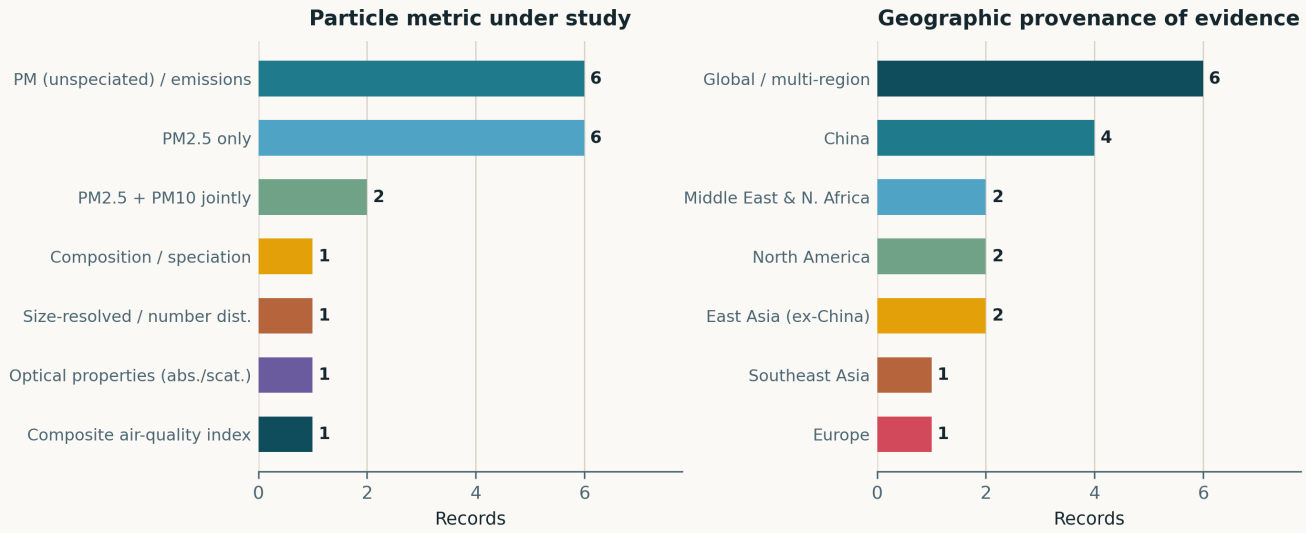


Fig. Left — particle metric: unspeciated PM/emissions and $PM_{2.5}$-only records dominate; one composition, one optical-property and one size-resolved record. Right — geography: China 4, not-stated or global 6, then Korea, East Asia, Iran, Turkiye, Thailand, Italy and the USA.

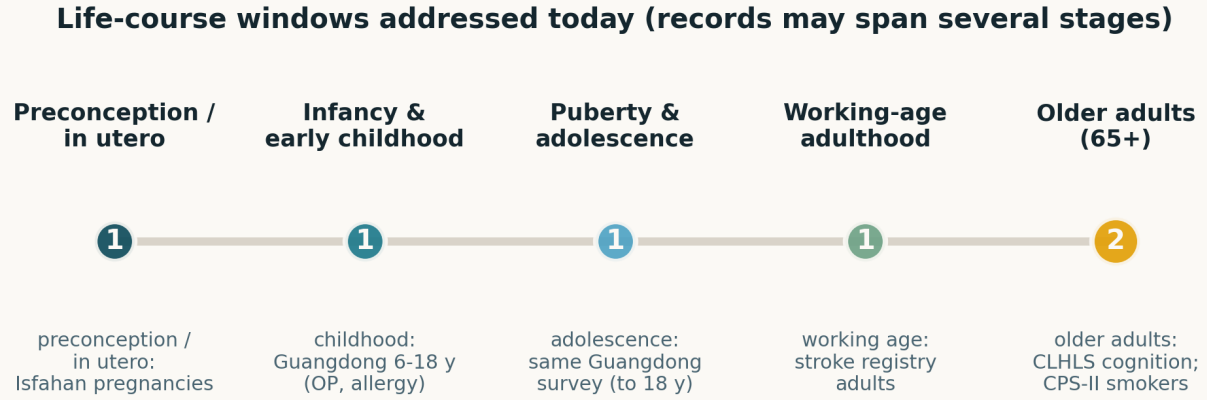


Fig. Life-course windows: every window populated by a single study except older adults (CLHLS cognition, CPS-II smokers). The childhood and adolescent counts are the same Guangdong survey.

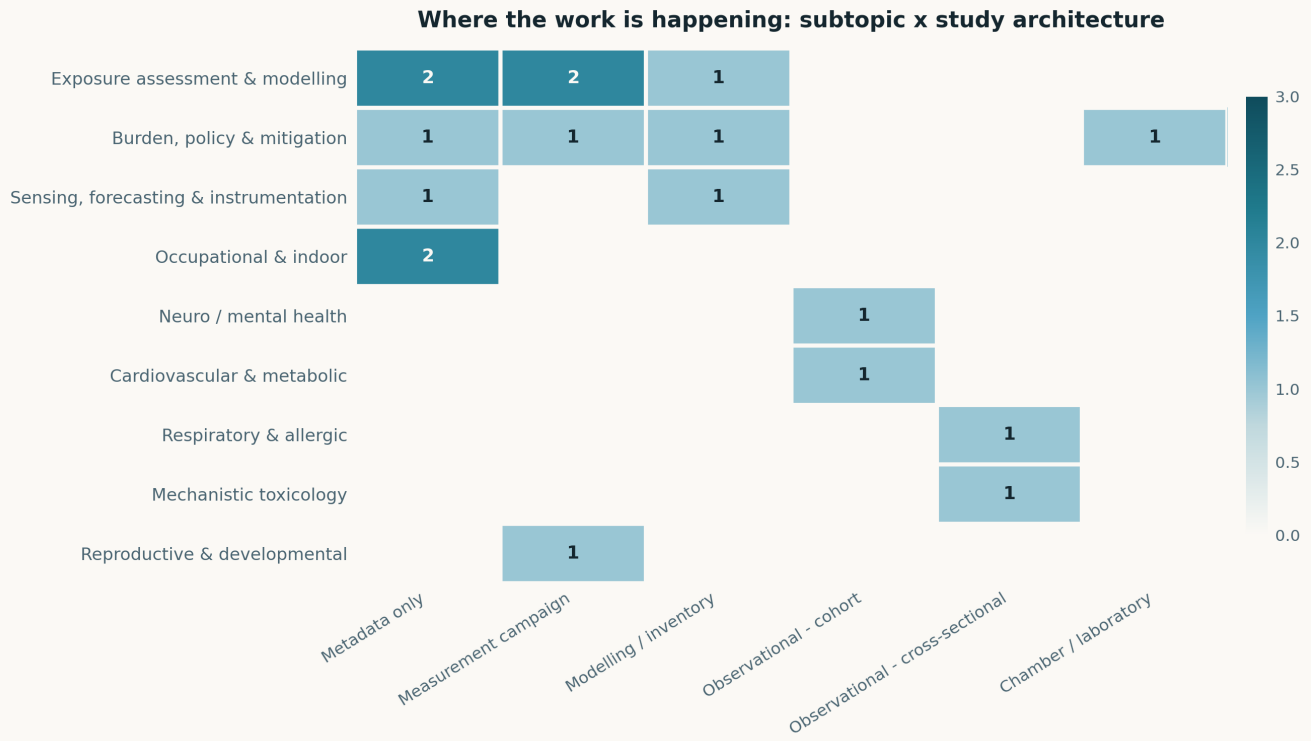


Fig. Subtopic × study architecture. The only cell above one is exposure assessment × measurement campaign; the metadata-only column spreads across sensing, burden and indoor bands.

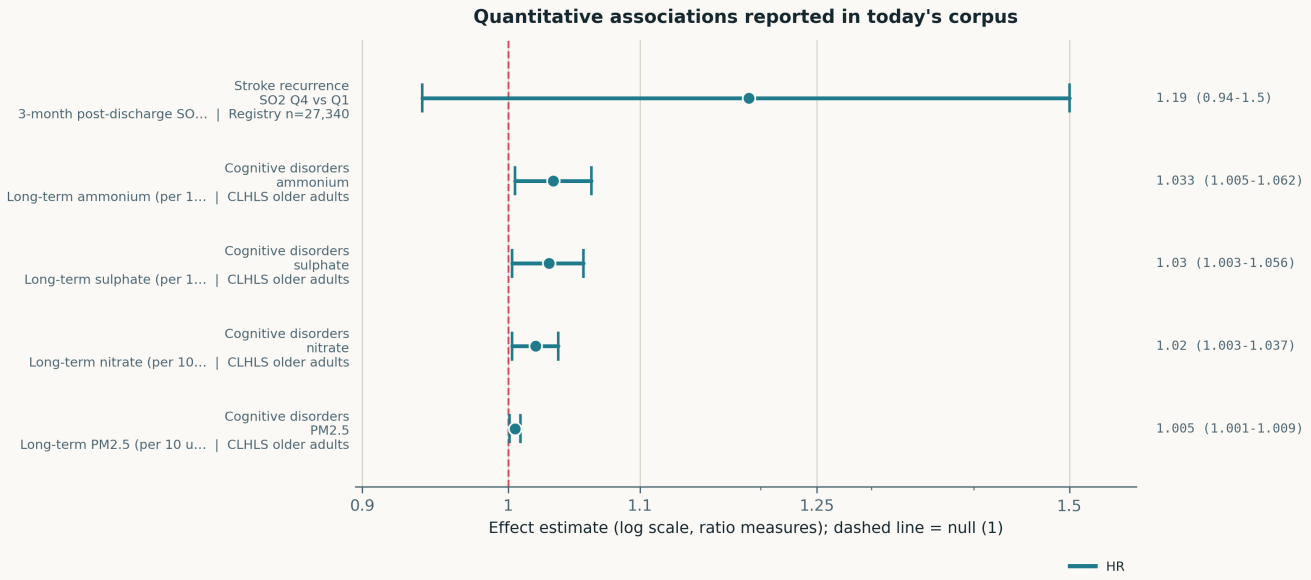


Fig. Ratio scale: four HRs for incident cognitive disorder per 10 μg m^{-3} from one Chinese cohort ($PM_{2.5}$, sulphate, nitrate, ammonium; single-pollutant models of collinear secondary components), and the quartile HR for post-discharge SO_2 and stroke recurrence from a Korean registry, whose CI crosses 1. No PM estimate from the stroke study is plotted because none was reported.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation 2026 J Glob Health

Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle

Chinese Longitudinal Healthy Longevity Study; Cox, Bayesian WQS and mediation models. Per 10 $\mu g\,m^{-3}$, $PM_{2.5}$ HR 1.005 (1.001-1.009), sulphate 1.030 (1.003-1.056), nitrate 1.020 (1.003-1.037), ammonium 1.033 (1.005-1.062), O_3 1.006; mixture OR 1.028 (1.006-1.040) with O_3 carrying the largest WQS weight. Favourable vs unfavourable lifestyle HR 0.488 (0.336-0.710); lifestyle mediated 8.4-9.5% of the pollutant associations. Limitation: an HR of 1.005 per 10 $\mu g\,m^{-3}$ is at the floor of detectability and the abstract gives neither n, case count nor how cognitive disorder was ascertained (MMSE cut-offs in CLHLS are education-sensitive); 'mediation' by lifestyle that is itself a confounder of residential exposure is not identifiable. Relevance: low - gridded component surfaces, no sensor component.

doi: 10.7189/jogh.16.04307

Cardiovascular & metabolic — 1 record

Lee CRCS-K post-discharge SO_2 stroke recurrence PM null 2026 J Stroke

Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke

27,340 acute ischaemic stroke patients (2014-2021), district-level IDW-interpolated pollutant means over the 3 months after discharge; cause-specific Cox with death as competing risk, cluster-robust SEs, splines. 764 recurrences, 542 deaths over a median 364.8 days. PM_{10} and $PM_{2.5}$ showed no association; only SO_2 did (overall spline p=0.024), with Q4 vs Q1 HR 1.19 (0.94-1.50), stronger in age >=75 and women (interaction p 0.051, 0.062). Limitation: district IDW from sparse monitors gives Berkson-type error that biases PM toward the null, and SO_2 concentrates in harbour cities (Incheon, Ulsan, Busan) - a place effect cannot be excluded. Relevance: moderate - a null PM result built on interpolation is exactly where denser networks change the answer.

doi: 10.5853/jos.2026.03265

Respiratory & allergic — 1 record

Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy 2026 Environ Sci Technol

Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China

131,346 children aged 6-18 in six Guangdong cities. $PM_{2.5}$ mass and DTT oxidative potential were positively associated with all three symptom outcomes; mass-normalised OP (DTTm) gave the most consistent associations, lower AIC/BIC and the greatest stability in two-pollutant models. Mixture analyses implicated several constituents; OP plausibly carried part of the Co, OC, Cu and Hg associations, the largest indirect share 36.2% for OC and rhinoconjunctivitis via DTTm. Limitation: cross-sectional symptom questionnaires; OP is presumably modelled or assigned from few filter sites, and the abstract reports no effect sizes, so the model-fit comparison cannot be weighed against exposure-error differences between metrics. Relevance: high - OP is not measurable by optical LCS, which bounds what a mass-only network can say about allergy.

doi: 10.1021/acs.est.6c05462

Reproductive & developmental — 1 record

Moradnia Isfahan pregnant urinary metals indoor dust AQI 2026 Environ Geochem Health

Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran

80 pregnant women; paired urine and vacuum dust analysed by ICP-OES. Dust metals ranked Pb>Cr>Ni>Cd (Pb mean 111.10 ug/g); urine Pb 10.52, Cr 8.30, Ni 4.14 ug/L. Urinary metals and AQI differed across urban zones (p<0.001); dust did not (Pb p=0.514). Building age correlated with dust Pb (rho 0.788). Dust and urinary profiles were uncorrelated. Limitation: n=80, spot urine without creatinine adjustment stated, and the only airborne metric is a zone-level AQI - no PM mass or PM-bound metal was measured, so 'air quality' enters only as an ecological covariate. Relevance: low - illustrates the gap that personal or indoor PM monitoring with filter speciation would fill.

doi: 10.1007/s10653-026-03508-x

Mechanistic toxicology — 1 record

Tang CPS-II plasma proteome air pollution lung cancer ever-smokers 2026 Environ Sci Technol

Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers

95 matched case-control pairs of ever-smokers with pre-diagnostic plasma proteomics from the CPS-II Nutrition Cohort. Seventeen proteins associated with both exposures and lung cancer (CX3CL1, ADAM15, CXCL17 among them, p<0.05); high-dimensional mediation (HIMA) flagged six, including CX3CL1 and CXCL17 (Bonferroni p<0.05). Limitation: 190 people against a high-dimensional proteome, and the mediation signals disappeared when cases diagnosed within a year of blood draw were excluded - consistent with reverse causation by subclinical tumour. The abstract does not name the pollutants' exposure model. Pathways (chemokine signalling, immune modulation) are hypothesis-generating only. Relevance: low - biomarker discovery, no sensing implication.

doi: 10.1021/acs.est.6c07407

Exposure assessment & modelling 5 records

Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia 2026 Atmos Chem Phys

Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations

Component-resolved LETKF assimilation in WRF-Chem of CALIOP-MODIS synergistic species-specific extinction profiles (dust, sea salt, BC, water-soluble), with spatial and observational constraints. The analysis reconstructs multi-layer vertical structure and corrects severe underestimation of elevated BC plumes, including persistent free-tropospheric BC over South Asia; AERONET and AD-Net lidar validation shows gains in AOT, extinction and single-scattering albedo. Limitation: CALIOP's narrow curtain and 16-day repeat leave most of the domain unobserved, the species classification is retrieval-model dependent, and the abstract gives no quantitative skill figures; surface $PM_{2.5}$ is not evaluated. Relevance: moderate - column and vertical constraints are what satellite-surface fusion with LCS networks currently lacks.

doi: 10.5194/acp-26-14671-2026

Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$ 2026 Environ Sci Technol

Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze

Winter-haze $PM_{2.5}$ from the North China Plain compared with the Sichuan Basin. Reduced-nitrogen NOCs were 40% of ESI+ organics. High-molecular-weight (m/z>250) Re-NOCs rose in humid versus dry haze and correlated with NH4+ and the aqueous-SOA fraction of water-soluble organics, alongside longer aliphatic chains and higher unsaturation - read as carbonyl-amino oligomerisation and cyclisation. A marker oligomer (C8H13N3O2) tracked WSON; stronger acidity and higher temperature favoured oligomerisation over cyclisation, enriching it in the Sichuan Basin. Limitation: ESI+ is semi-quantitative and ionisation-biased, number of samples is not stated, and mechanism is inferred from correlation. Relevance: low - brown-carbon NOCs add absorbing mass that optical scattering sensors misweight under humid haze.

doi: 10.1021/acs.est.6c09940

Lu street-view greenery street canyon PM concentrations 2026 Environ Sci: Atmos

Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery

Street-view imagery quantified street greenery quantity and configuration, which were related to PM concentrations in street canyons; the authors attribute the associations to low-level vegetation barriers restricting ventilation. The record's abstract is two sentences: no city, sample size, PM metric, measurement method or effect direction and magnitude are given. Limitation: on what is available, the inference is correlational and the proposed ventilation mechanism is untested; treat as near-metadata until the full text is read. Relevance: moderate - siting of street-level sensors relative to hedges and low canopy is a known representativeness issue for LCS networks, and image-derived greenery is a cheap covariate for spatial calibration.

doi: 10.1039/d6ea00086j

He NYC-METS size-resolved organic aerosol offline AMS 2026 Aerosol Sci Technol

Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS

METADATA ONLY. Title and authors only; no abstract in Crossref, Europe PMC or Semantic Scholar at harvest. The record reports size-resolved organic aerosol characterised by offline aerosol mass spectrometry - re-aerosolised filter or impactor extracts - within the NYC-METS campaign in New York City. Sampling period, size cuts, source apportionment and sample counts cannot be stated. Limitation: offline AMS loses water-insoluble and semi-volatile fractions in extraction, so size-resolved OA mass is a lower bound; no inference beyond the title is possible. Relevance: moderate - size-resolved organic composition governs refractive index and hygroscopicity, the two terms optical LCS corrections approximate.

doi: 10.1080/02786826.2026.2746516

Barnett-Itzhaki hourly compound heat air pollution extremes 2026 Atmos Pollut Res

Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title claims that hourly data expose compound heat-air-pollution extreme windows that daily means hide. Pollutants, location, thresholds, record length and how many events are missed by daily averaging cannot be stated. Limitation: unverifiable beyond the title; compound-extreme definitions are threshold-dependent, and hourly co-occurrence of heat and O_3 is expected by photochemistry, so the PM-specific contribution is unknown. Relevance: moderate - sub-daily exposure windows are precisely what continuous low-cost monitors resolve and daily regulatory filters do not.

doi: 10.1016/j.apr.2026.103243

Sensing, forecasting & instrumentation 2 records

Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO_2 2026 Atmosphere

Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul

17,520-h panel (2022-2023) pairing six bridge traffic sensors with the nearest monitors. A Prophet traffic-only baseline gave negative R2 for every bridge x pollutant. LightGBM, XGBoost and a BiLSTM-attention model with meteorology, calendar, precipitation lags and short autoregressive terms reached mean hold-out R2 ~0.77 (0.58-0.86), RMSE down 47-63%; the three families were practically equivalent. SHAP ranked autocorrelation, season, wind direction and heavy vehicles above total traffic. Limitation: autoregressive pollutant terms make this nowcasting persistence, not a traffic proxy - the headline R2 largely reflects lagged concentrations, and 'closest monitor' distance is unstated. Relevance: moderate - shows traffic counts alone do not substitute for measurement in complex coastal flow.

doi: 10.3390/atmos17101024

Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors 2026 Build Environ

Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title describes an urban $PM_{2.5}$ prediction model with implicit physics constraints and uncertainty quantification, plus driving-factor attribution. Architecture, city, input features, horizon, skill metrics and whether the uncertainty is calibrated cannot be stated. Limitation: 'implicit' physics constraints in ML usually mean loss-function regularisation, which does not guarantee physical consistency out of distribution; nothing beyond the title is verifiable. Relevance: moderate - calibrated predictive uncertainty is what allows LCS-fed forecasts to propagate sensor error honestly.

doi: 10.1016/j.buildenv.2026.115449

Occupational & indoor 2 records

Suh residential occupant activities airborne microplastic emissions 2026 Build Environ

Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title reports experimental quantification of microplastic particle emissions from occupant activities in residential spaces and a relative-importance analysis of generation factors (activity type, textiles, surfaces presumably). Size range, identification method (FTIR, Raman, pyrolysis GC-MS), emission rates and whether the inhalable fraction was measured cannot be stated. Limitation: unverifiable beyond the title; microplastic counts are method-bound and seldom expressed as PM mass. Relevance: moderate - indoor activity-resolved particle sources are what household LCS see as unexplained spikes; polymer fibres also scatter differently from combustion aerosol.

doi: 10.1016/j.buildenv.2026.115399

Du classroom CO_2 proxy PM VOC ventilation types 2026 Indoor Environments

CO_2 as a proxy for PM and VOCs in classrooms across ventilation types and activities

METADATA ONLY. Title and authors only; no abstract retrievable at harvest. The title tests whether CO_2 tracks PM and VOCs in classrooms across ventilation types and activities - i.e. whether a cheap NDIR sensor can stand in for particle and VOC monitoring. Number of classrooms, ventilation categories, PM instrumentation and agreement statistics cannot be stated. Limitation: unverifiable beyond the title; physically, CO_2 tracks occupant-generated bioeffluents and air exchange, whereas PM indoors also depends on outdoor infiltration, resuspension and filtration, so decoupling under mechanical ventilation is expected. Relevance: high - directly asks whether school CO_2 monitor rollouts can be read as PM exposure surveillance.

doi: 10.1016/j.indenv.2026.100200

Burden, policy & mitigation 4 records

Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG 2026 Atmosphere

Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City

Harmonised 2024 integrated inventory plus enterprise statistics for Jiaxing; 54 non-zero source categories scored on an air-pollutant H-score (mean of $PM_{2.5}$, NOx, SO_2, VOCs, NH_3 shares) against a GWP100 CO_2-eq G-score, screened at the 70th percentile. Coupled co-control candidates: CHP, thermal power, motor vehicles; decoupled single-pollutant priorities: industrial solvents (VOCs) and paved-road dust (PM). A 75th-percentile benchmark implies 7,941 t VOC reduction among reporting enterprises. Limitation: equal-weight averaging of pollutant shares ignores toxicity and secondary PM formation potential; no ambient or health validation. Relevance: low - road-dust PM priority is the coarse fraction optical LCS undercount.

doi: 10.3390/atmos17101023

Warminski SRC pellets boiler ESP PM CO emissions 2026 Molecules

Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions

Short-rotation coppice pellets (willow, poplar, black locust) versus pine and birch in a pellet boiler with an electrostatic precipitator. Species significantly affected PM and CO; the ESP cut mean PM by 52.5% (53.81 to 25.54 mg/m3) without affecting CO. Highest PM: willow without ESP (90.96 mg/m3); lowest: pine with ESP (17.51). Poplar gave low PM but the highest CO; black locust the best balance (combustion efficiency 93.20%). Limitation: one boiler, total PM only (no size distribution or PM_1 fraction, where residential-combustion mass sits), and steady-state operation that misses start-up peaks. Relevance: low - source-side mitigation; a 52% ESP removal at total-mass level likely overstates sub-micron removal.

doi: 10.3390/molecules31193443

Brunori Rome urban vineyard grapevine leaf PM interception 2026 Plants (Basel)

Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars

Urban vineyard of 170 cultivars at Rome's Botanical Garden, 2025 season: 54 cultivars screened by leaf micromorphology, five assessed in depth (anatomy, chlorophyll, reflectance indices, SEM-counted PM). After ~20 days without heavy rain, leaf-surface PM ranged from 52,024 (Barbera) to 77,462 particles/mm2 (Maturano), comparable to some urban tree species; adaxial deposition was over twice abaxial. Limitation: particle counts, not mass or size-resolved deposition flux; five cultivars, one site, one accumulation window, and no ambient PM measurement to convert counts into a removal rate. Relevance: low - green-infrastructure mitigation claims need co-located ambient PM, which a low-cost sensor at canopy height would supply cheaply.

doi: 10.3390/plants15193014

Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes 2026 Atmos Pollut Res

Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai (2021-2024)

METADATA ONLY. Title and author only; no abstract retrievable at harvest. The title states a distributed-lag GAM comparing the sensitivity of $PM_{2.5}$ to burning-control measures in an urban regime (Bangkok) and a biomass-burning-dominated regime (Chiang Mai), 2021-2024. Effect sizes, lag structure, the burning-control variable (fire counts, policy dates) and confounder set cannot be stated. Limitation: observational policy attribution in a GAM is sensitive to meteorological adjustment and transboundary smoke from Myanmar and Laos; single author. Relevance: moderate - Chiang Mai is a dense PMS-sensor setting (DustBoy network), and regime-specific response is where LCS spatial coverage tests policy effects.

doi: 10.1016/j.apr.2026.103236

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Zhang CLHLS $PM_{2.5}$ components cognitive disorders lifestyle mediation	Association between long-term ambient air pollution exposure and cognitive disorders among older adults in China: the mediating role of lifestyle	*J Glob Health*	Prospective cohort	Long-term $PM_{2.5}$, sulphate, nitrate, ammonium, O3 (per 10 $\mu g/m^3$)	China (CLHLS, multi-province)
Cardiovascular & metabolic					
Lee CRCS-K post-discharge SO2 stroke recurrence PM null	Post-Discharge Ambient Sulfur Dioxide Exposure and Stroke Recurrence After Acute Ischemic Stroke	*J Stroke*	Prospective cohort (registry linkage)	3-month post-discharge district PM_{10}, $PM_{2.5}$, NO2, SO2, CO, O3 (IDW)	South Korea (CRCS-K-NIH registry)
Reproductive & developmental					
Moradnia Isfahan pregnant urinary metals indoor dust AQI	Spatial variability of heavy metals in pregnant women: linking urinary biomarkers, indoor dust contamination, and air quality in Isfahan, Iran	*Environ Geochem Health*	Cross-sectional environmental survey	Zone-level Air Quality Index; indoor settled dust metals (not airborne PM)	Iran (Isfahan)
Respiratory & allergic					
Wang Guangdong $PM_{2.5}$ oxidative potential DTT childhood allergy	Associations of $PM_{2.5}$ and Its Oxidative Potential with Childhood Allergy-Related Symptoms: A Cross-Sectional Study in Guangdong Province, China	*Environ Sci Technol*	School-based cross-sectional	$PM_{2.5}$ mass, oxidative potential (DTTv, DTTm), metals, carbonaceous, ions	China (Guangdong, six cities)
Mechanistic toxicology					
Tang CPS-II plasma proteome air pollution lung cancer ever-smokers	Analysis of the Potential Mediating Role of Plasma Proteins in the Association Between Air Pollution Exposure and Lung Cancer Risk Among Ever-Smokers	*Environ Sci Technol*	Nested case-control	Six ambient pollutants and their mixture (pre-diagnosis)	USA (CPS-II Nutrition Cohort)
Occupational & indoor					
Du classroom CO2 proxy PM VOC ventilation types	CO2 as a proxy for PM and VOCs in classrooms across ventilation types and activities	*Indoor Environments*	Metadata only (no abstract)	Classroom PM (fractions not stated) vs CO2 and VOCs	Not stated in record
Suh residential occupant activities airborne microplastic emissions	Microplastics emissions from occupant activities in residential spaces: Experimental quantification and relative importance analysis of particle generation factors	*Build Environ*	Metadata only (no abstract)	Airborne microplastic particles from occupant activities	Not stated in record
Sensing, forecasting & instrumentation					
Birinci Bosphorus bridge traffic ML proxy $PM_{2.5}$ NO2	Machine-Learning Proxy of Air-Quality Variability Associated with Intercontinental Bridge Traffic over the Bosphorus in Istanbul	*Atmosphere*	Machine-learning model	Hourly $PM_{2.5}$, PM_{10}, coarse fraction, NO2, NO, NOx	Turkiye (Istanbul)
Ren physics-constrained UQ urban $PM_{2.5}$ prediction driving factors	Integrating Implicit Physics Constraints and Uncertainty Quantification for Urban $PM_{2.5}$ Prediction and Driving Factor Analysis	*Build Environ*	Metadata only (no abstract)	Urban $PM_{2.5}$ (prediction target)	Not stated in record
Exposure assessment & modelling					
Barnett-Itzhaki hourly compound heat air pollution extremes	Hourly Resolution Reveals Hidden Temporal Windows of Compound Heat-Air Pollution Extremes Masked by Daily Averaging	*Atmos Pollut Res*	Metadata only (no abstract)	Hourly air pollution (pollutant set not stated) co-occurring with heat	Not stated in record
Cheng LETKF WRF-Chem CALIOP component-resolved aerosol assimilation East Asia	Constraining the vertical profiles of aerosol component and heating rate over east Asia through assimilating CALIOP observations	*Atmos Chem Phys*	Chemical transport modelling	Species-resolved extinction profiles: dust, sea salt, black carbon, water-soluble	East Asia
He NYC-METS size-resolved organic aerosol offline AMS	Characteristics of size-resolved organic aerosols using offline aerosol mass spectrometry: NYC-METS	*Aerosol Sci Technol*	Metadata only (no abstract)	Size-resolved organic aerosol (offline AMS)	USA (New York City)
Lu street-view greenery street canyon PM concentrations	Association between urban greenery and particulate matter concentrations in street canyons: a local-scale study based on street view imagery	*Environ Sci: Atmos*	Field measurement	Street-canyon PM concentrations (metric not stated in abstract)	Not stated in abstract

First author	Focus	Journal	Design	Metric	Region
Xiao NCP Sichuan haze reduced-nitrogen NOC oligomerisation $PM_{2.5}$	Aqueous Carbonyl-Amino Aggregation Drives Atmospheric Nitrogen-Containing Oligomer Formation in China Haze	*Environ Sci Technol*	Field measurement	$PM_{2.5}$ nitrogen-containing organic compounds (ESI+ HRMS)	China (North China Plain, Sichuan Basin)
Burden, policy & mitigation					
Brunori Rome urban vineyard grapevine leaf PM interception	Plant-Atmosphere Interactions in Urban Vineyards of Rome: A Digital Phenotyping Approach for the Functional Selection of Grapevine Cultivars	*Plants (Basel)*	Field measurement	Leaf-surface PM particle counts (SEM, particles/mm2)	Italy (Rome)
Polwiang Bangkok Chiang Mai burning control DLNM GAM $PM_{2.5}$ regimes	Differential sensitivity of urban and biomass-dominated $PM_{2.5}$ regimes to burning control: A distributed-lag generalized additive model analysis of Bangkok and Chiang Mai	*Atmos Pollut Res*	Metadata only (no abstract)	Daily $PM_{2.5}$ (urban vs biomass-dominated regimes)	Thailand (Bangkok, Chiang Mai)
Warminski SRC pellets boiler ESP PM CO emissions	Short-Rotation Coppice Pellets as Biofuels in a Modern Boiler with an Electrostatic Precipitator: Particulate Matter and Carbon Monoxide Emissions	*Molecules*	Laboratory measurement	Flue-gas total PM mass (mg/m3) with and without ESP	Not stated in abstract
Zhang Jiaxing inventory co-control screening $PM_{2.5}$ GHG	Inventory-Based Co-Control Screening of Air Pollutants and Greenhouse Gases in a Typical Yangtze River Delta City	*Atmosphere*	Emission inventory	Inventory $PM_{2.5}$ (with NOx, SO2, VOCs, NH3) vs CO2-eq	China (Jiaxing, Yangtze River Delta)

Provenance and method

Entry window **19 October 2026**, single day (Monday). Harvester legs: **PubMed** health **4**, sensing **1**; **Europe PMC 96** (16 AGRICOLA retro-index dropped); **Crossref by ISSN 55**; **arXiv** 1 entry dated 19 October; **OpenAlex** HTTP 429. 137 unique, 50 already carried, 87 fresh. **PubMed connector** (EDAT 19 Oct) 6 PMIDs, 2 appended — **89** candidates screened, **19 admitted**, **70 rejected** with logged reasons; 1 admitted record **held** (arXiv DOI not yet registered), leaving **18**. **Consensus** calibration sweep: 10 hits, 0 new. **DOI gate** 0 fail / 0 warn after the hold. **Trial watch**: 0 registrations updated 19–20 October. Abstract retry (PubMed, Europe PMC, Crossref, Semantic Scholar) recovered none of the six day-0 Elsevier and T&F records; **6 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.